



Grübler's Formula, Task Space and Workspace

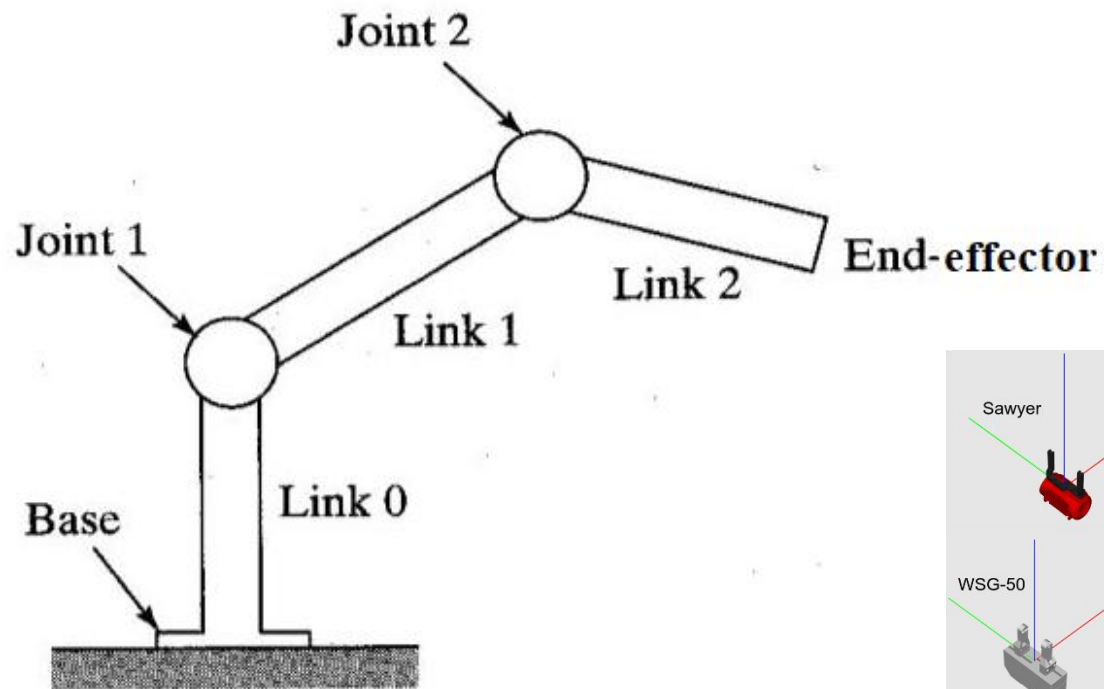
CS 6341 Robotics

Instructor Yu Xiang

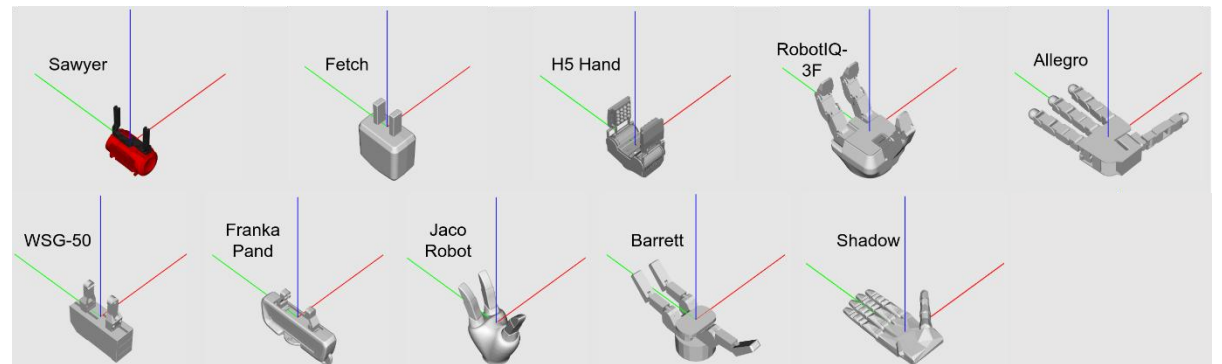
The University of Texas at Dallas

Robot Mechanisms

- Links and Joints

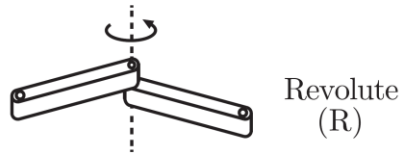


Franka Emika

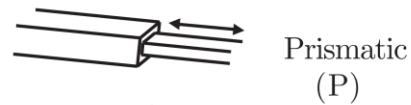


End-effectors

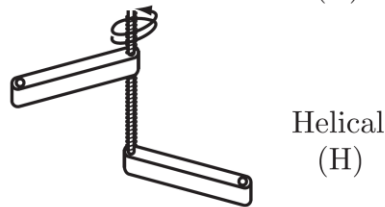
Degrees of Freedom of Robot Joints



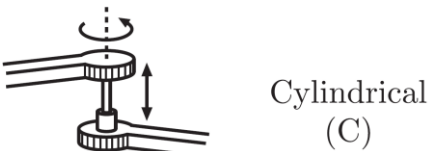
Revolute (R)



Prismatic (P)



Helical (H)



Cylindrical (C)



Universal (U)



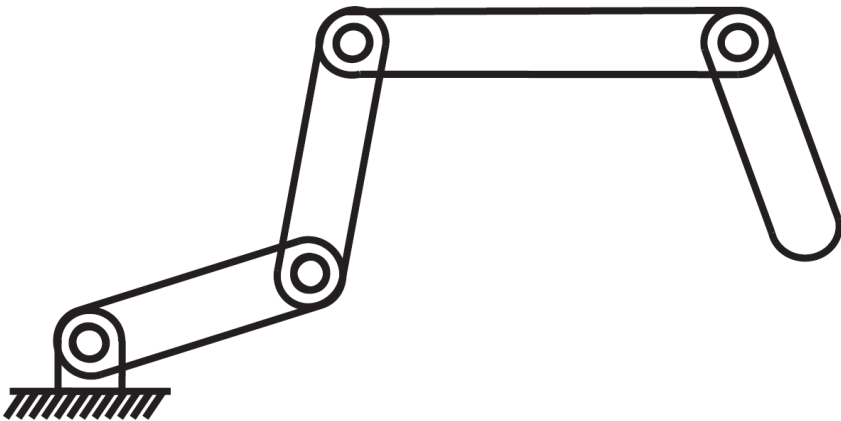
Spherical (S)

Joint type	dof f	Constraints c between two planar rigid bodies	Constraints c between two spatial rigid bodies
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	N/A	5
Cylindrical (C)	2	N/A	4
Universal (U)	2	N/A	4
Spherical (S)	3	N/A	3

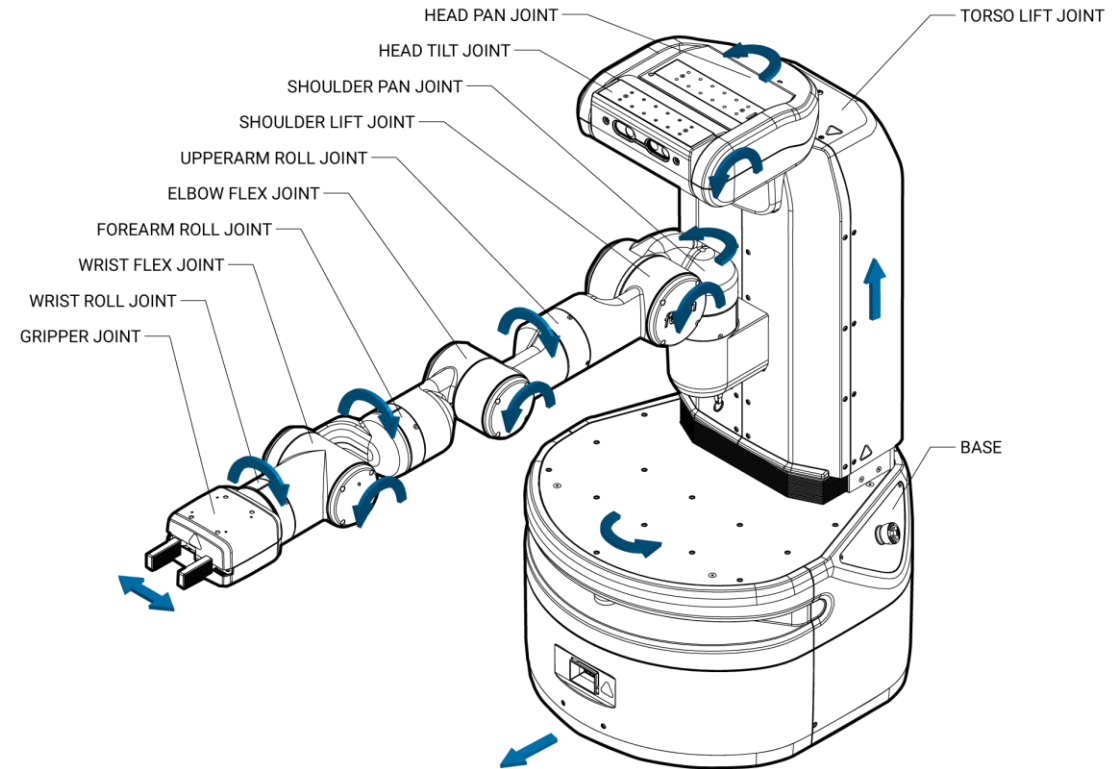
Planar rigid body: 3 DOF

Spatial rigid body: 6 DOF

Degrees of Freedom of a Robot



- 4 revolute joints
- 4 DOFs



- 7 revolute joints for the arm
- 7 DOFs

Grübler's Formula

- The number of degrees of freedom of a mechanism with links and joints can be calculated using Grübler's formula

$$\text{degrees of freedom} = (\text{sum of freedoms of the bodies}) - (\text{number of independent constraints})$$

- Consider the following setting
 - A robot with N links, J joints (consider ground as one link)
 - Each link has m DOF (planar link? spatial link?)
 - Number of freedoms by joint i f_i
 - Number of constraints by joint i c_i

$$f_i + c_i = m$$

Grübler's Formula

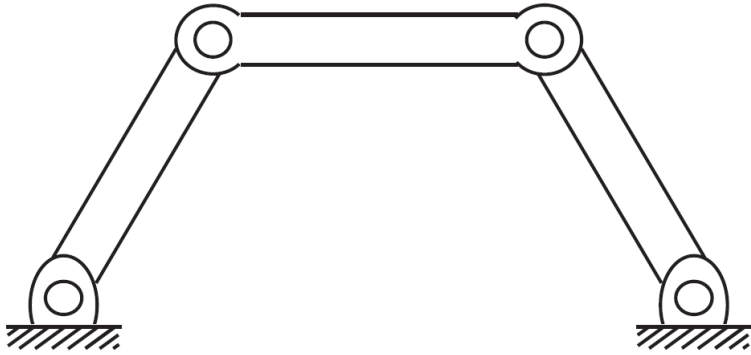
$$\text{dof} = \underbrace{m(N-1)}_{\text{rigid body freedoms}} - \underbrace{\sum_{i=1}^J c_i}_{\text{joint constraints}} \quad \text{Ground is regarded as a link}$$

$$= m(N-1) - \sum_{i=1}^J (m - f_i)$$

$$= m(N-1-J) + \sum_{i=1}^J f_i.$$

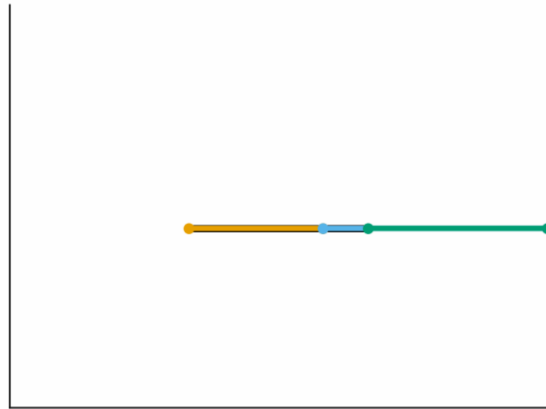
Assume all joint constraints are independent.

Grübler's Formula



The planar four-bar linkage

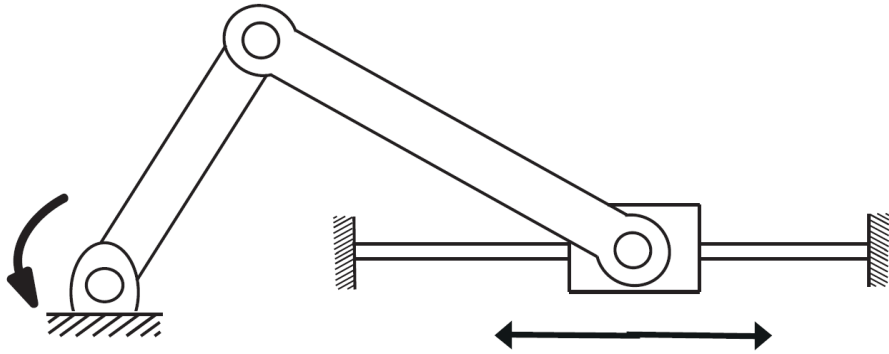
Four-Bar Linkage (Crank-Rocker Mechanism)



- How many links?
 - 4 (one is ground)
- Each link has m DOF. What is m ?
 - $m=3$

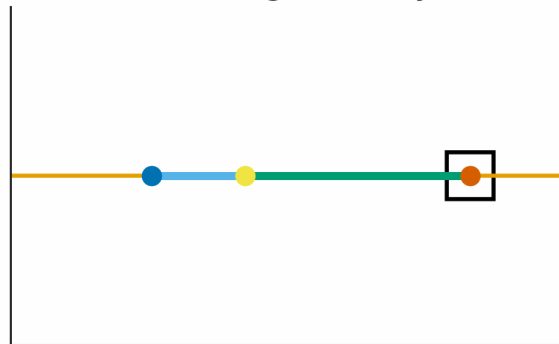
$$\begin{aligned}\text{DOF} &= m(N - 1 - J) + \sum_{i=1}^J f_i \\ &= 3(4 - 1 - 4) + \sum_{i=1}^4 1\end{aligned}$$

Grübler's Formula



Slider-crank mechanism (planar)

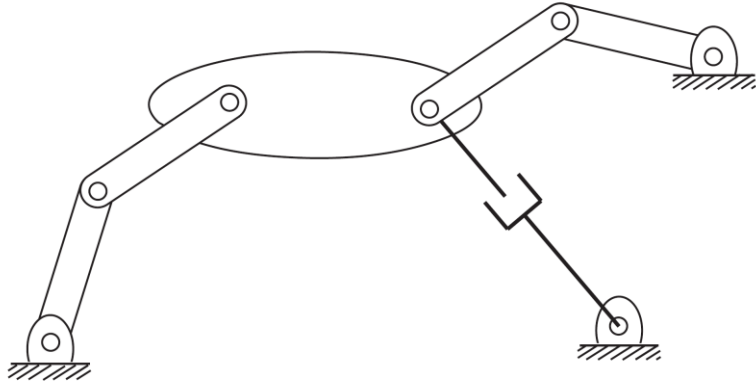
Slider-Crank Robot Linkage (Rotary → Linear Motion)



- How many links?
 - 4 (one is ground)
- Each link has m DOF. What is m ?
 - $m=3$
- How many joints?
 - 3 revolute joints, 1 prismatic joint

$$\begin{aligned} \text{DOF} &= m(N - 1 - J) + \sum_{i=1}^J f_i \\ &= 3(4 - 1 - 4) + \sum_{i=1}^4 1 \end{aligned}$$

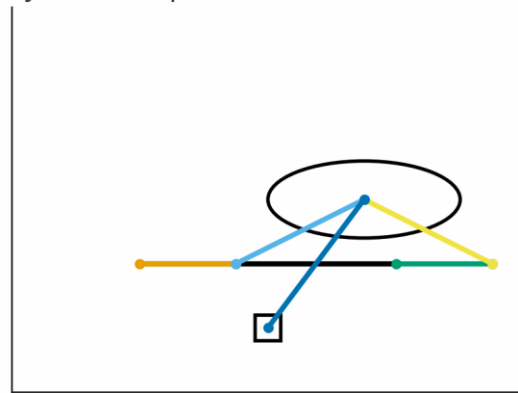
Grübler's Formula



A planar mechanism with two overlapping joints

- How many links?
 - 8 (one is ground)
- Each link has m DOF. What is m ?
 - $m=3$
- How many joints?
 - 8 revolute joints, 1 prismatic joint

Hybrid Manipulator with Oval Platform

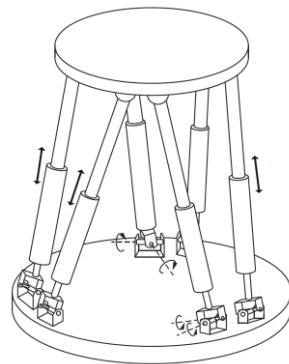
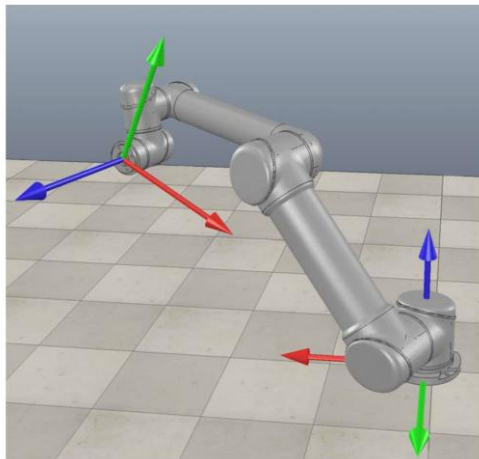


$$\text{dof} = 3(8 - 1 - 9) + 9(1) = 3$$

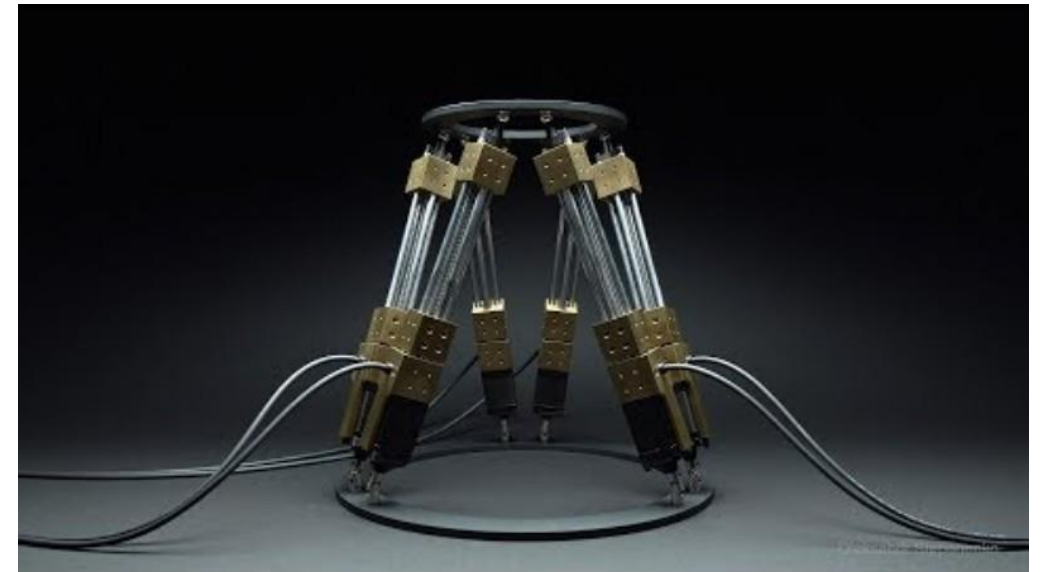
Read more examples in the textbook Lynch & Park

Open-Chain vs. Closed-Chain

- Open-chain mechanisms: without a closed loop
- Closed-chain mechanisms: with a closed loop
- Examples
 - A person standing with both feet



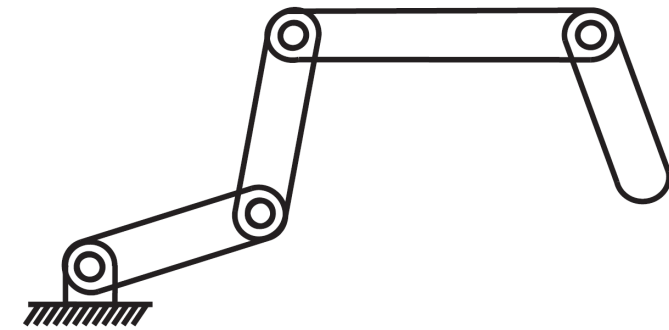
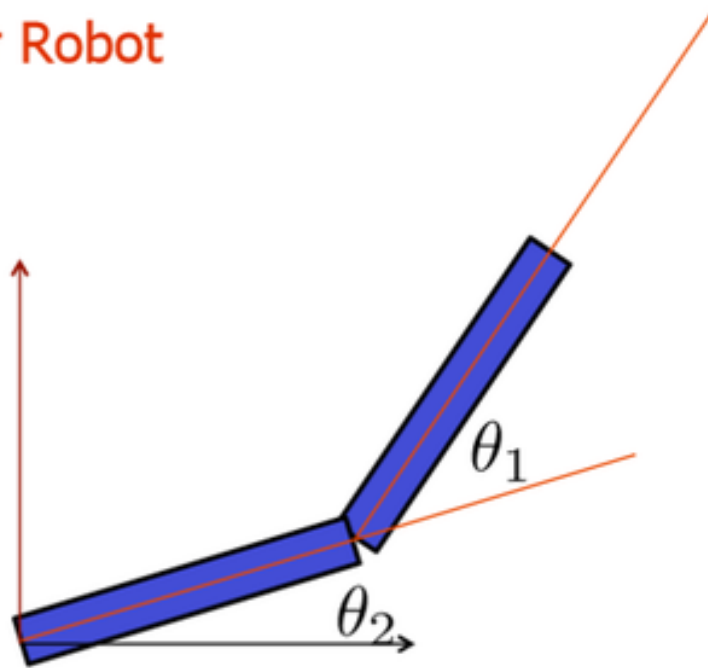
Stewart-Gough platform



<https://www.youtube.com/watch?v=xiECumcaEx0>

Configuration Space of a Robot

Two Link Planar Robot



- 4 revolute joints
- 4 DOFs

Configuration Space Topology

- Configuration specifies the position of a robot
- For a robot with n joints, the configuration is a vector in $\mathbb{R}^n$
 - C-space
- Joints may have limits, upper bound and lower bound
- Topology: shape of the space
 - Consider all the feasible points in the configuration space

Configuration Space Topology

- n-dimensional Euclidean space $\mathbb{R}^n$
- n-dimensional sphere in a (n+1)-dimensional Euclidean space S^n
 - Two-dimensional surface of a sphere in three-dimensional space S^2



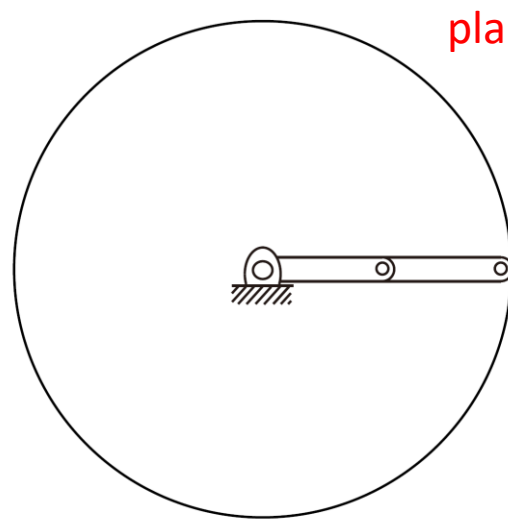
S^2

Task Space

- The task space is a space in which the robot's task can be naturally expressed
- Task examples
 - Draw on a piece of paper: $\mathbb{R}^2$
 - Manipulate a rigid body: C-space of the rigid body
- Task space is driven by the task, independently of the robot

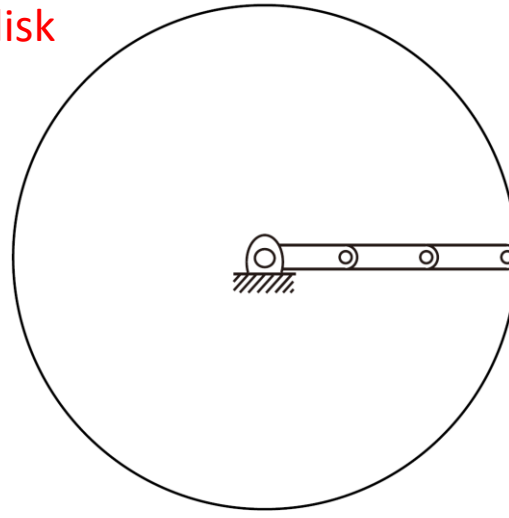
Workspace

- The workspace is a specification of the configurations that **the end-effector of the robot can reach**.
- Depends on the robot structure, independent of the task

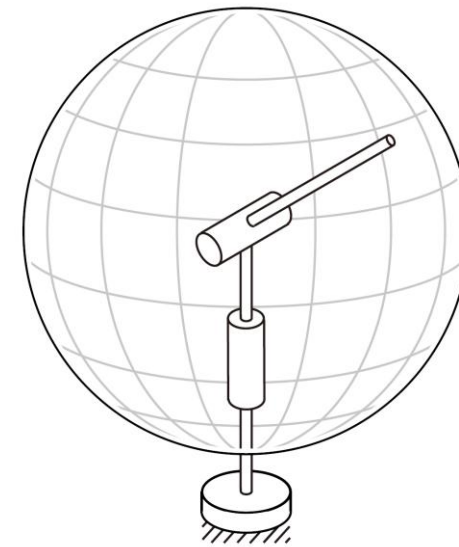


planar disk

a planar 2R open chain



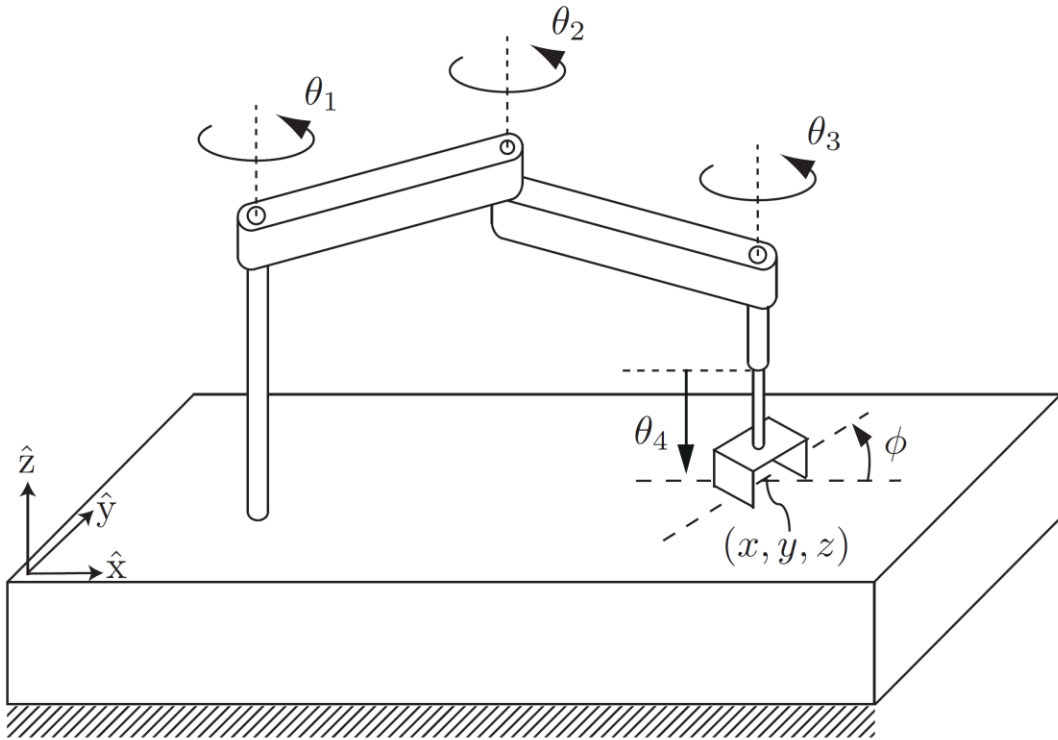
a planar 3R open chain



Surface of
a sphere

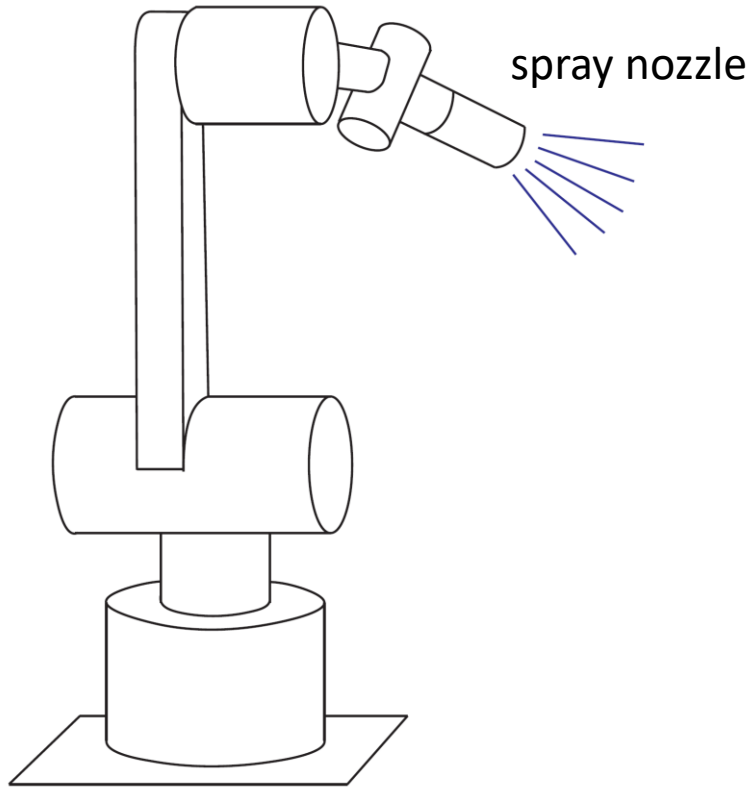
a spherical 2R open chain

SCARA Robot



- End-effector configuration (x, y, z, ϕ)
- Task space $\mathbb{R}^3 \times S^1$
- Workspace
 - Reachable (x, y, z, ϕ)

A 6R Robot



A spray-painting robot

- End-effector configuration

$$(x, y, z) \quad (\theta, \phi)$$

Cartesian position of
the nozzle

Spherical coordinates to describe
the direction in which the nozzle
is pointing

- Task space $\mathbb{R}^3 \times S^2$

- Workspace

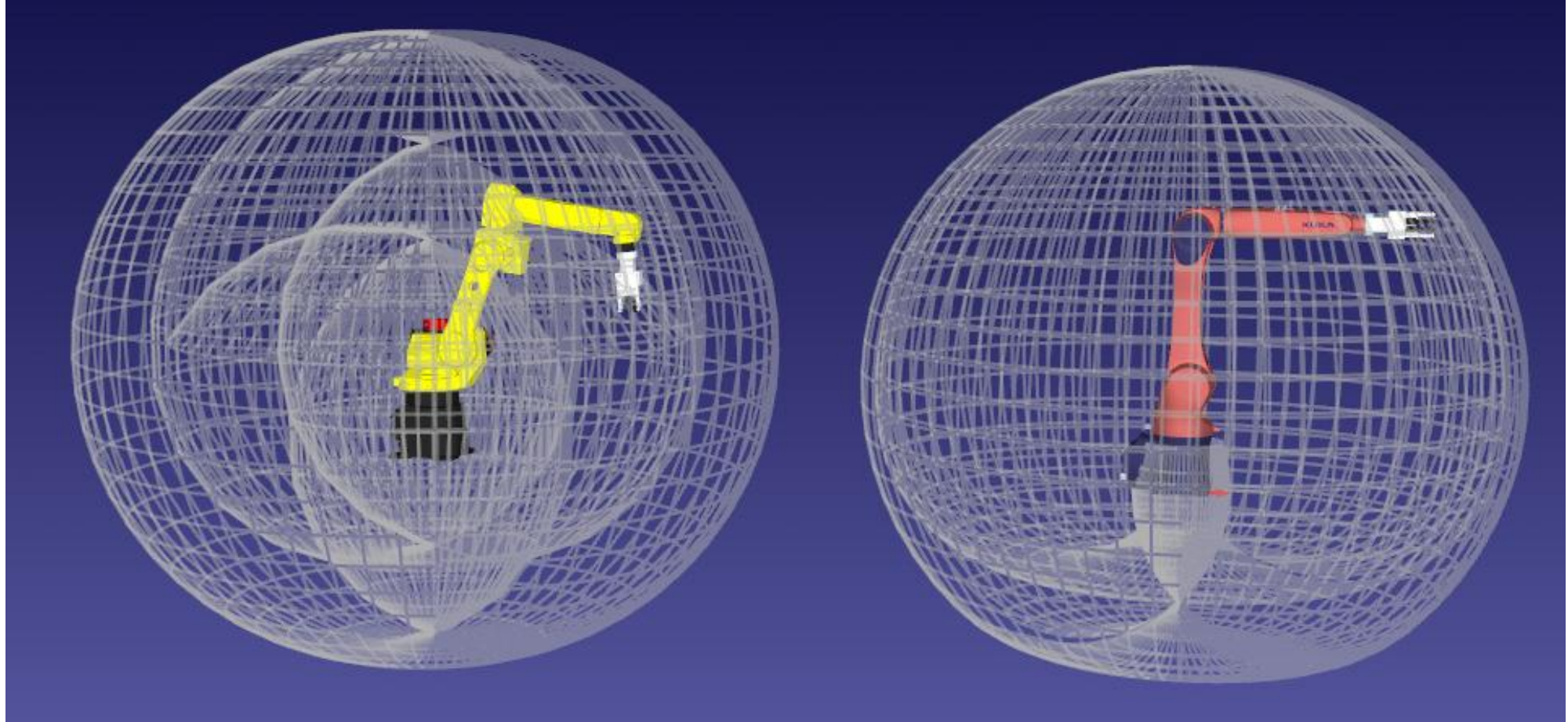
- Reachable

$$(x, y, z) \quad (\theta, \phi)$$



<https://www.rnaautomation.com/case-study/robotic-spray-booth/>

Workspace Examples



<https://robodk.com/blog/robot-workspace-visualization/>

Summary

- Gröbler's Formula
- Task space
- Workspace

Further Reading

- Chapter 2 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017
<http://hades.mech.northwestern.edu/images/7/7f/MR.pdf>
- T. Lozano-Perez. Spatial planning: a configuration space approach. A.I. Memo 605, MIT Artificial Intelligence Laboratory, 1980.
<http://people.csail.mit.edu/tlp/>
- W. M. Boothby. An Introduction to Differentiable Manifolds and Riemannian Geometry. Academic Press, 2002.